

AquaTracker - Module Overview

This document describes the purpose and function of each major module used in the AquaTracker project.

Modules and Their Purpose

Arduino Uno: Main controller that reads sensor values, displays them on the OLED, and sends commands to the GSM module.

GSM Module: Enables SMS communication so the system can alert the user when water quality or water level becomes critical.

OLED Display: Shows real-time pH, water level, quality percentage, and status information locally.

pH Sensor: Measures the acidity or alkalinity of the water to monitor water condition.

Temperature Sensor: Measures the water temperature for environmental monitoring.

Turbidity Sensor: Measures water quality by detecting suspended particles and cloudiness in the water.

Ultrasonic Sensor: Measures the distance to the water surface to estimate the tank water level.

Power Supply: Provides stable power to the Arduino, sensors, OLED display, and GSM module for reliable operation.

Breadboard / Wiring: Provides the physical connection setup between all components and allows prototyping and testing.